

$$\begin{array}{ccc}
1_{\mathcal{D}} & \xrightarrow{\varphi_0} & F1_{\mathcal{C}} \\
j_{FX} \downarrow & & \downarrow Fj_X \\
[FX, FX]_{\mathcal{D}} & \xleftarrow{\varphi_{X,X}} & F[X, X]_{\mathcal{C}}
\end{array}$$